CHRISTIAN ENGELS

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narfinger.github.io narfinger DBLP https://dblp.org/pers/hd/e/Engels:Christian
Google Scholar https://scholar.google.com/citations?user=8TIu-IsAAAAJ&hl=de

ABOUT MYSELF

I am a programmer currently working in academic research in algorithms and complexity theory. I am looking for interesting programming job opportunities. As you can see best from my github, I am proficient in multiple languages (please notice that some repositories include multiple languages, as I use them in to learn new concepts.) My knowledge of datastructures and problem solving is advanced. In the academic environment, producing results is highly based on collaboration, communication and, of course, problem solving. The below mentioned DBLP link, gives all my current published research. I believe you can see my success in these skills.
Recently, I started experimenting with unreal engine and unity and I am learning fast how to adapt my knowledge to game programming. If you have any questions, feel free to reach out to me.

BIGGEST PROJECTS

Viola: A music player written in Rust with a frontend of GTK. It supports playlists and smartplaylists. Over time, I moved the frontend to React/Typescript with the backend in Rust. I am currently in the process of transforming the frontend to Rust with seed-rs and Web Assembly (http://github.com/Narfinger/viola).

Open Source Contributions: Small contributions to Open Source Projects such as jellyfin (C#), jellyfin-webos (Javascript) and Clementine (C++)

Theorygrabber: Fetching new Theoretical Computer Science Papers from arxiv.org and eccc.org. Parsing rss and html feeds, downloading the papers and uploading them to google drive. Written in Rust (http://github.com/Narfinger/TheoryGrabber).

Toggl-rs: A library and CLI to interface with www.toggl.com written in Rust (http://github.com/Narfinger/toggl-rs).

GENERAL INFORMATION

- Date of Birth: 13.08.1985.
- Nationality: German.
- Fluent in: German, English.

EDUCATION AND EMPLOYMENT

PostDoc
National Institute of Technology (Big Data Mathematics Project)
2019.10 - Tokyo, Japan

PostDoc
Indian Institute of Technology Bombay
2017.08 - 2019.09 Mumbai, India

PostDoc
Kyoto University
2017.04 - 2017.09 Kyoto, Japan

PostDoc
Tokyo Institute of Technology (Exploring the Limits of Computation Project)
2015.11 - 2017.03 Tokyo, Japan

PhD. in Computer Science **Thesis:** Why are certain polynomials hard? **Advisor:** Prof. Markus Blaeser
Saarland University
2010.07 - 2015.11 Saarbruecken, Germany

MSc. In Computer Science **Thesis:** Randomness Efficient Identity Testing s. **Advisor:** Prof. Markus Blaeser

Saarland University

2008 - 2010 Saarbruecken, Germany

BSc. in Computer Science **Thesis:** Probabilistic Analysis of Algorithms for the TSP. **Advisor:** Dr. Bodo Manthey

Saarland University

2005 - 2008 Saarbruecken, Germany

PUBLICATIONS

Journal Articles

- Engels, Christian, Mohit Garg, et al. (2020). "On Expressing Majority as a Majority of Majorities". In: *SIAM J. Discret. Math.* 34.1, pp. 730–741. DOI: 10.1137/18M1223599. URL: https://doi.org/10.1137/18M1223599.
- Engels, Christian and B. V. Raghavendra Rao (2020). "On hard instances of non-commutative permanent". In: *Discret. Appl. Math.* 277, pp. 127–138. DOI: 10.1016/j.dam.2019.09.003. URL: https://doi.org/10.1016/j.dam.2019.09.003.
- Bläser, Markus and Christian Engels (2019b). "Parameterized Valiant's Classes". In: *CoRR* abs/1907.12287. URL: http://arxiv.org/abs/1907.12287.
- Chillara, Suryajith et al. (2018b). "A Near-Optimal Depth-Hierarchy Theorem for Small-Depth Multilinear Circuits". In: *CoRR* abs/1804.02520. URL: http://arxiv.org/abs/1804.02520.
- – (2018c). "A Near-Optimal Depth-Hierarchy Theorem for Small-Depth Multilinear Circuits". In: *Electronic Colloquium on Computational Complexity (ECCC)* 25, p. 62. URL: https://eccc.weizmann.ac.il/report/2018/062.
- Engels, Christian (2016a). "Dichotomy Theorems for Homomorphism Polynomials of Graph Classes". In: *J. Graph Algorithms Appl.* 20.1, pp. 3–22. DOI: 10.7155/jgaa.00382. URL: https://doi.org/10.7155/jgaa.00382.
- Engels, Christian, B. V. Raghavendra Rao, and Karteek Sreenivasaiah (2016). "Lower Bounds and Identity Testing for Projections of Power Symmetric Polynomials". In: *Electronic Colloquium on Computational Complexity (ECCC)* 23, p. 153. URL: http://eccc.hpi-web.de/report/2016/153.
- Bringmann, Karl et al. (2015). "Random Shortest Paths: Non-Euclidean Instances for Metric Optimization Problems". In: *Algorithmica* 73.1, pp. 42–62. DOI: 10.1007/s00453-014-9901-9. URL: https://doi.org/10.1007/s00453-014-9901-9.
- Engels, Christian (2014). "Dichotomy Theorems for Homomorphism Polynomials of Graph Classes". In: *CoRR* abs/1412.0423. URL: http://arxiv.org/abs/1412.0423.
- Engels, Christian and B. V. Raghavendra Rao (2014). "New Algorithms and Hard Instances for Non-Commutative Computation". In: *CoRR* abs/1409.0742. URL: http://arxiv.org/abs/1409.0742.
- Bringmann, Karl et al. (2013a). "Random Shortest Paths: Non-Euclidean Instances for Metric Optimization Problems". In: *CoRR* abs/1306.3030. URL: http://arxiv.org/abs/1306.3030.
- Engels, Christian and Bodo Manthey (2009). "Average-case approximation ratio of the 2-opt algorithm for the TSP". in: *Oper. Res. Lett.* 37.2, pp. 83–84. DOI: 10.1016/j.orl.2008.12.002. URL: https://doi.org/10.1016/j.orl.2008.12.002.

Conference Proceedings

- Bläser, Markus and Christian Engels (2019a). "Parameterized Valiant's Classes". In: *14th International Symposium on Parameterized and Exact Computation, IPEC 2019, September 11-13, 2019, Munich, Germany*. Vol. 148, 3:1–3:14. DOI: 10.4230/LIPIcs.IPEC.2019.3. URL: https://doi.org/10.4230/LIPIcs.IPEC.2019.3.
- Chillara, Suryajith et al. (2018a). "A Near-Optimal Depth-Hierarchy Theorem for Small-Depth Multilinear Circuits". In: *59th IEEE Annual Symposium on Foundations of Computer Science, FOCS 2018, Paris, France, October 7-9, 2018*, pp. 934–945. DOI: 10.1109/FOCS.2018.00092. URL: https://doi.org/10.1109/FOCS.2018.00092.

- Engels, Christian, B. V. Raghavendra Rao, and Karteek Sreenivasaiah (2017). “On \varSigma \wedge \varSigma \wedge \varSigma Circuits: The Role of Middle \varSigma Fan-In, Homogeneity and Bottom Degree”. In: *Fundamentals of Computation Theory - 21st International Symposium, FCT 2017, Bordeaux, France, September 11-13, 2017, Proceedings*. Vol. 10472, pp. 230–242. DOI: 10.1007/978-3-662-55751-8_19. URL: https://doi.org/10.1007/978-3-662-55751-8%5C_19.
- Engels, Christian and B. V. Raghavendra Rao (2016). “On Hard Instances of Non-Commutative Permanent”. In: *Computing and Combinatorics - 22nd International Conference, COCOON 2016, Ho Chi Minh City, Vietnam, August 2-4, 2016, Proceedings*. Vol. 9797, pp. 171–181. DOI: 10.1007/978-3-319-42634-1_14. URL: https://doi.org/10.1007/978-3-319-426 5C_14.
- Engels, Christian (2015). “Dichotomy Theorems for Homomorphism Polynomials of Graph Classes”. In: *WALCOM: Algorithms and Computation - 9th International Workshop, WALCOM 2015, Dhaka, Bangladesh, February 26-28, 2015. Proceedings*. Vol. 8973, pp. 282–293. DOI: 10.1007/978-3-319-15612-5_25. URL: https://doi.org/10.1007/978-3-319-15612 5C_25.
- Bringmann, Karl et al. (2013b). “Random Shortest Paths: Non-euclidean Instances for Metric Optimization Problems”. In: *Mathematical Foundations of Computer Science 2013 - 38th International Symposium, MFCS 2013, Klosterneuburg, Austria, August 26-30, 2013. Proceedings*. Vol. 8087, pp. 219–230. DOI: 10.1007/978-3-642-40313-2_21. URL: https://doi.org/10.1007/978-3-642-40313-2%5C_21.
- Engels, Christian, Bodo Manthey, and B. V. Raghavendra Rao (2012). “Random Shortest Path Metrics with Applications”. In: *11th Cologne-Twente Workshop on Graphs and Combinatorial Optimization, Munich, Germany, May 29-31, 2012. Extended Abstracts*, pp. 121–124.
- Bläser, Markus and Christian Engels (2011). “Randomness Efficient Testing of Sparse Black Box Identities of Unbounded Degree over the Reals”. In: *28th International Symposium on Theoretical Aspects of Computer Science, STACS 2011, March 10-12, 2011, Dortmund, Germany*. Vol. 9, pp. 555–566. DOI: 10.4230/LIPIcs.STACS.2011.555. URL: https://doi.org/10.4230/LIPIcs.STACS.2011.555.